PhotoMind: A Multimodal Personal Photo Knowledge Retrieval System

Course: Prompt Engineering — Building Agentic Systems
Assignment: Take-Home Final — Reinforcement Learning for Agentic AI Systems
Platform: CrewAI (Python) · PyTorch
Domain: Personal Productivity
RL Approaches: Contextual Bandits (Thompson Sampling + UCB) + DQN Confidence Calibration

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1. System Overview

PhotoMind turns a personal photo library into a queryable knowledge base. The system addresses a genuine gap: smartphone users accumulate thousands of photos — receipts, bills, food, screenshots, documents — but cannot search them by meaning or extract facts from them using natural language.

Core capabilities: - Analyze photos using GPT-4o Vision to extract text (OCR), entities, and semantic descriptions - Build a persistent structured knowledge base from the analysis results - Answer natural-language queries with three distinct retrieval strategies - Return confidence-graded answers with source photo attribution - Gracefully decline queries the knowledge base cannot answer

Real-world use cases demonstrated: - "How much did I spend at ALDI?" → finds all 5 ALDI receipts, aggregates total with source photos - "Show me photos of pizza" → finds food photos matching the description - "What type of food do I photograph most?" → analyzes patterns across all 25 photos - "What was my electric bill?" → correctly declines (no such photo in the library)

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2. System Architecture

2.1 High-Level Architecture

```
                        PhotoMind


       INGESTION PIPELINE              QUERY PIPELINE
      (Process.sequential)         (Process.hierarchical)

     [Photo Analyst]               [Controller Agent]
                                    (Manager/Orchestrator)

       Scan
      photos/                     [Retriever] [Synthesizer



       Analyze
      GPT-4o V.



       Index → JSON KB

     [Knowledge Retriever]



                Knowledge Base (photo_index.json)
     { photos: [ { filename, image_type, ocr_text,
                   description, entities, confidence } ] }
```

2.2 Data Flow

Ingestion:

```
photos/ directory

      (DirectoryReadTool)
List of image files (JPG, PNG, HEIC)

      (PhotoVisionTool → GPT-4o Vision API)
Per-photo analysis: { image_type, ocr_text, description, entities, confidence }

      (FileReadTool + direct write)
knowledge_base/photo_index.json  [25 photos indexed]
```

Query:

```
User natural-language query
```

```
     (Controller classifies intent)
Query type: factual | semantic | behavioral

     (PhotoKnowledgeBaseTool)
Search results with relevance scores

     (Insight Synthesizer)
Structured answer: { answer, confidence_grade, source_photos, reasoning }
```

2.3 Memory Architecture

Both pipelines use CrewAI's built-in memory with a **local sentence-transformer model** (`all-MiniLM-L6-v2`) for vector embeddings. This stores agent interactions in a local ChromaDB vector store, enabling contextual awareness across agent steps without requiring external embedding API access.

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3. Agent Roles and Responsibilities

PhotoMind uses four specialized agents across two crews.

3.1 Controller Agent (Query Crew — Manager)

Property	Value
Role	PhotoMind Controller
Process	Manager in hierarchical crew
Delegation	`allow_delegation=True`

Responsibility: Receives raw user queries, classifies intent (factual / semantic / behavioral), and delegates to the appropriate specialist agent. Acts as the chief librarian — decides *which* search strategy will best answer the question before any retrieval happens.

Decision logic: The controller understands that "how much was my bill?" (factual) requires entity extraction, "photos that feel like summer" (semantic) requires description matching, and "what food do I photograph most?" (behavioral) requires corpus-wide aggregation.

Backstory: *"You are the chief librarian of a personal photo knowledge base. You classify first, then delegate. If you are unsure, you say so."*

3.2 Photo Analyst (Ingestion Crew)

Property	Value
Role	Photo Analyst
Tools	PhotoVisionTool, DirectoryReadTool

Property	Value
Process	Sequential (steps 1 and 2 of ingestion)

Responsibility: Scans the `photos/` directory, then passes each image to GPT-4o Vision for analysis. Extracts structured knowledge: image classification, OCR text, semantic description, and named entities (amounts, dates, vendors, locations). Never fabricates information not visible in the image.

Key design decision: This agent is explicit about confidence — when text is unclear (blurry receipt, low-light photo), it reports partial extraction with a lower confidence score rather than guessing.

3.3 Knowledge Retriever (Both Crews)

Property	Value
Role	Knowledge Retriever
Tools	PhotoKnowledgeBaseTool (custom), FileReadTool, JSONSearchTool (agent-level)
Process	Sequential (step 3 of ingestion), subordinate in query crew

Responsibility: In ingestion: writes the final knowledge base JSON file. In queries: executes the search against the knowledge base using the appropriate strategy, returns ranked results with evidence. Note: the query task restricts this agent to only `PhotoKnowledgeBaseTool` via task-level tool override, preventing fallback searches that produce misleading weak matches.

Backstory: *"You are a research librarian with perfect recall. When no good match exists, you say so clearly rather than returning a weak match disguised as confident."*

3.4 Insight Synthesizer (Query Crew)

Property	Value
Role	Insight Synthesizer
Tools	FileReadTool, SerperDevTool (optional)
Process	Subordinate in hierarchical query crew

Responsibility: Takes raw retrieval evidence and synthesizes a grounded, human-readable answer. Applies three strict rules: (1) every claim cites a source photo, (2) every answer includes a confidence grade A–F, (3) if evidence is ambiguous, say so explicitly. May use web search (SerperDevTool) to enrich context when the user's `SERPER_API_KEY` is set.

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4. Tool Integration and Functionality

4.1 Built-in Tools

Tool	Agent	Purpose	Configuration
`DirectoryReadTool`	Photo Analyst	Scans `photos/` directory, lists all image files with extensions JPG/PNG/HEIC/WebP	`directory=settings.photos_directory`
`FileReadTool`	Knowledge Retriever, Insight Synthesizer	Reads the knowledge base JSON file; Synthesizer uses it to re-read source photos for context	Default, unrestricted path
`SerperDevTool`	Insight Synthesizer	Optional web search enrichment — adds public context (e.g., restaurant info, product details) when a relevant Serper API key is present	Graceful fallback: not added if key missing

Note: `JSONSearchTool` is configured on the Knowledge Retriever agent but is **not** available during query tasks. The query task explicitly restricts the retriever to only `PhotoKnowledgeBaseTool` to prevent fallback searches that produce weak, misleading matches. This was a deliberate design choice after observing that fallback tools caused the system to return irrelevant results instead of properly declining unanswerable queries.

Mapping to assignment requirements:

Required Category	Tool(s)	Justification
Web search or data retrieval	`DirectoryReadTool`, `SerperDevTool`	Directory scanning retrieves the photo inventory; web search enriches answers with public context
Data processing or transformation	`JSONSearchTool`	Embedding-based semantic search over the JSON knowledge base using sentence-transformer vectors
Communication or output formatting	`FileReadTool`	Provides agents structured access to the knowledge base for assembling cited, formatted answers

Tool selection rationale: - `DirectoryReadTool` is the natural fit for directory enumeration — it handles recursive listing and returns structured file metadata - `FileReadTool` provides safe, agent-

readable access to the knowledge base without exposing raw Python file I/O - `SerperDevTool` enriches behavioral answers (e.g., "I photographed Patel Brothers 3 times" could be enriched with store location context)

4.2 Custom Vision Tool (`PhotoVisionTool`)

Located at `src/tools/photo_vision.py`. Wraps GPT-4o Vision for image analysis.

Key implementation details: - Registers `pillow_heif` opener at import time, enabling transparent HEIC image loading - Converts images to JPEG in-memory via base64 (required by OpenAI vision API) - Guards against HEIC RGBA/YCbCr modes: `if img.mode not in ("RGB", "L"): img.convert("RGB")` - Returns structured JSON with keys: `image_type`, `ocr_text`, `description`, `entities`, `confidence` - On error: returns error string (not raise) so agents surface failures gracefully

4.3 Custom Knowledge Base Tool

Documented in full in Section 5.

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5. Custom Tool Documentation

PhotoKnowledgeBaseTool

File: `src/tools/photo_knowledge_base.py`
Class: `PhotoKnowledgeBaseTool(BaseTool)`
Input schema: `PhotoKBQueryInput(BaseModel)`

Purpose The core differentiating component of PhotoMind. Implements a three-strategy retrieval system with automatic query-intent routing, confidence scoring, and source attribution. Without this tool, the system could only do generic semantic search. With it, factual extraction ("how much?"), visual similarity ("photos that feel like summer"), and behavioral analysis ("what do I photograph most?") all work correctly.

Inputs

Parameter	Type	Default	Description
`query`	`str`	required	Natural language question about the user's photos
`query_type`	`str`	`"auto"`	Force a strategy: `"factual"`, `"semantic"`, `"behavioral"`, or `"auto"`
`top_k`	`int`	`3`	Number of results to return
`confidence_threshold`	`float`	`0.15`	Minimum score to include a result

Outputs


```
{
  "query_type_detected": "factual",
  "results": [
    {
      "photo_id": "uuid",
      "photo_path": "photos/IMG_1853.HEIC",
      "relevance_score": 0.55,
      "evidence": "vendor: ALDI; OCR text match; amounts: 18.69",
      "image_type": "receipt"
    }
  ],
  "confidence_grade": "B",
  "confidence_score": 0.55,
  "answer_summary": "Best match: photos/IMG_1853.HEIC (type: receipt, confidence: B). Evidence
  "source_photos": ["photos/IMG_1853.HEIC"],
  "warning": null
}
```


Query Intent Classification Rule-based classification using keyword lists. Behavioral signals are checked before factual signals to prevent over-triggering on shared keywords:

```
Query → _classify_query()
    Contains "how many", "most", "pattern", "breakdown" → behavioral
    Contains "how much", "date", "address", "vendor", "items" → factual
    Default → semantic
```

Search Strategies **Factual Search** — Targets receipts, bills, and documents with specific extractable facts: 1. Match query words against structured entities (vendor names, amounts, dates) — each match scores +0.4 2. Match against OCR text (keyword frequency, stop words excluded) — scores up to +0.5 3. Boost if image_type matches query context (+0.2) 4. Include all amount entities in evidence (e.g., `amounts: $18.69, $2.15`) so the LLM can compute totals 5. **Aggregation mode:** queries containing "how much", "spend", "total", etc. return all matching results instead of `top_k`, and the summary includes an aggregated total across receipts

Semantic Search — Targets photos by visual meaning or mood: 1. Apply `_clean()` (lowercase + strip punctuation) to both query and descriptions 2. Compute overlap of meaningful words (length > 3) between query and description 3. Normalize by meaningful query word count (not total — stop words excluded) 4. Formula: `len(meaningful_overlap) / max(len(meaningful_query_words), 1) × 0.8` 5. Image-type boost: if query mentions image type (e.g., "document") +0.2

Behavioral Search — Targets questions about patterns across the whole corpus: 1. Aggregate `image_type` distribution across all photos 2. Aggregate entity value frequencies (most common vendors, items) 3. Return representative photos from the dominant pattern 4. Summary includes full distribution stats

Confidence Grading

Grade	Score Range	Meaning
A	0.7	Strong match — high-confidence answer
B	0.5	Good match — reliable for most purposes
C	0.35	Moderate — answer likely correct, verify if critical
D	0.2	Weak — returned best available, may be wrong
F	< 0.2	No reliable match — system declines to answer

Error Handling

- KB file not found → returns structured error JSON, confidence F
- Corrupted JSON → catches `JSONDecodeError`, returns error JSON
- Empty knowledge base → returns error JSON with guidance to run ingestion
- All errors return strings (not exceptions) so agents can report them to users

Limitations

- Semantic search uses keyword overlap, not true embeddings — misses synonyms
- Behavioral analysis aggregates by `image_type` field only, not by semantic clusters
- Confidence calibration is empirical — thresholds tuned on a 25-photo corpus

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6. Orchestration Design

6.1 Ingestion Crew — Sequential Process

```
Task 1: Scan          → Photo Analyst + DirectoryReadTool
Task 2: Analyze       → Photo Analyst + PhotoVisionTool  [context: scan results]
Task 3: Index         → Knowledge Retriever + FileReadTool [context: analyze results]
```

The sequential process ensures each task can pass its output as `context` to the next. The `create_analyze_task` and `create_index_task` tasks receive upstream outputs via `context=[previous_task]`, enabling the pipeline to flow structured data forward without requiring agents to re-read files.

Idempotency: The scan task checks `photo_index.json` for already-indexed filenames and skips them. Re-running ingestion on a partially indexed corpus only processes new photos.

6.2 Query Crew — Hierarchical Process with Planning

```
Manager: Controller Agent

      Task 1 → Knowledge Retriever: searches KB with PhotoKnowledgeBaseTool
      Task 2 → Insight Synthesizer: synthesizes answer with confidence grade [context: Task 1]
```

The query pipeline uses two chained tasks: 1. **Retrieval Task** — assigned to Knowledge Retriever, uses only `PhotoKnowledgeBaseTool` (task-level tool override restricts the agent to this single tool, preventing fallback searches that produce weak matches). Returns raw results with relevance scores and evidence. 2. **Synthesis Task** — assigned to Insight Synthesizer, receives the retrieval

results via `context=[query_task]`. Produces a structured JSON answer with `confidence_grade`, `source_photos`, `query_type`, and `reasoning`.

Key crew settings: - `process=Process.hierarchical` — manager controls all delegation - `manager_agent=controller` — explicit manager assignment - `planning=True` — the crew generates a plan before executing, improving task decomposition - `memory=True` — cross-step context retention via sentence-transformer embeddings

Why hierarchical? Query answering benefits from a manager that can decide when to re-query with different parameters, when to ask for web enrichment, and when to synthesize a "decline" response. Sequential would force a fixed path; hierarchical allows adaptive decision-making.

6.3 Feedback Loop — Adaptive Confidence Thresholds

```
Evaluation Run → FeedbackStore (knowledge_base/feedback_store.json)

      Per-strategy accuracy: factual / semantic / behavioral
      Adaptive adjustment: accuracy < 70% → +0.05 threshold (more conservative)
                           accuracy  90% → -0.05 threshold (less conservative)
      Applied in: PhotoKnowledgeBaseTool._run() → adjusts confidence_threshold per query
```

The `FeedbackStore` (`src/tools/feedback_store.py`) implements a persistent feedback loop: - After each eval query, the outcome (correct/incorrect, strategy used, confidence score) is recorded - Per-strategy accuracy rates are computed once 3+ samples exist - Adaptive confidence threshold adjustments are stored and applied by `PhotoKnowledgeBaseTool` on subsequent queries - Eval run history is tracked in `eval/results/eval_history.json` for trend analysis

This ensures the system learns from evaluation results: strategies with low accuracy become more conservative (higher threshold → fewer false positives), while high-accuracy strategies become more permissive.

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7. Challenges and Solutions

Challenge 1: Python Version Incompatibility

Problem: CrewAI requires Python < 3.14. macOS default was 3.14.x.
Solution: Used pyenv to create a venv with Python 3.10.14: `~/.pyenv/versions/3.10.14/bin/python3 -m venv .venv`

Challenge 2: HEIC Image Support

Problem: 14 of 25 photos were HEIC (iPhone format). PIL does not decode HEIC natively.
Solution: Added `pillow-heif` and called `register_heif_opener()` at module import. Added an `img.convert("RGB")` guard for HEIC files decoded in RGBA/YCbCr mode.

Challenge 3: Free-Tier Gemini API Quota Exhaustion

Problem: Attempted to use Gemini free tier (gemini-2.5-flash, 20 RPD limit). Multiple failed debug runs exhausted the 20 daily requests within hours.

Solution: Switched to OpenAI GPT-4o with a paid API key. Implemented a direct ingestion script (`ingest_direct.py`) to minimize API calls: 1 call per photo instead of ~4.

Challenge 4: OpenAI Embeddings API Blocked

Problem: `memory=True` in Crew defaults to OpenAI's embedding API (`text-embedding-3-small`). The provided API key didn't have embeddings access.
Solution: Configured sentence-transformers as the local embedding provider: `embedder={"provider": "sentence-transformer", "config": {"model": "all-MiniLM-L6-v2"}}`. This runs entirely locally, no API calls required.

Challenge 5: Punctuation Breaking Query Matching

Problem: Query `"How much at ALDI?"` failed because `"aldi?"` (with `?`) didn't match entity `"aldi"` in the knowledge base.
Solution: Added `_clean()` helper applying `re.sub(r'[^\w\s]', '', text.lower())` to all text before comparison.

Challenge 6: Semantic Search Scoring Too Strict

Problem: `"Show me photos of pizza"` scored `1/5 × 0.8 = 0.16` — below the threshold — because stop words ("show", "me", "of") inflated the denominator.
Solution: Normalized by meaningful word count only (words with length > 3), giving `1/3 × 0.8 = 0.27` — above the adjusted threshold of 0.15.

Challenge 7: Evaluation Case-Sensitivity Bug

Problem: Eval lowercased all text then compared against mixed-case expected filenames (`"IMG_1853.HEIC"` vs `"img_1853.heic"`). All retrieval was marked wrong.
Solution: One-line fix: `tc["expected_photo"].lower() in parsed["source_photos"]`.

Challenge 8: Context Accumulation Causing TPM Errors

Problem: Reusing one Crew instance for 20 eval queries caused CrewAI memory to accumulate ~32k tokens by query 19, exceeding the 30k TPM limit.
Solution: Instantiate a fresh Crew per query in the eval harness.

Challenge 9: Stop Words Causing False Matches in Factual Search

Problem: `"What was my electric bill this month?"` returned grade D with irrelevant photos because stop words "what" and "this" (length > 3, passing the filter) matched against unrelated OCR text.
Solution: Added a `_STOP_WORDS` set and excluded these words from OCR text matching. Electric bill query now correctly returns grade F with zero results.

Challenge 10: Agents Ignoring Tool Decline Signals

Problem: When `PhotoKnowledgeBaseTool` returned grade F with no results for unanswerable queries, the CrewAI agents used fallback tools (`JSONSearchTool`, `FileReadTool`) to find irrelevant weak matches and fabricated explanations like "file encoding errors."

Solution: Applied task-level tool override (`tools=[PhotoKnowledgeBaseTool(...)]`) on the query task, restricting the retriever to only the primary search tool. Combined with a directive decline message in the tool output and strengthened agent instructions.

Challenge 11: Aggregation Queries Truncated by top_k

Problem: `"How much did I spend at ALDI?"` only returned 3 of 5 ALDI receipts due to the `top_k=3` limit, and the evidence lacked dollar amounts for the LLM to compute a total.
Solution: Added aggregation detection for queries containing "how much", "spend", "total", etc. These queries now return all matching results with amount entities included in the evidence, and the summary includes a computed aggregated total.

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8. System Performance Analysis

8.1 Evaluation Setup

- **Test set:** 20 hand-labeled queries across 4 categories using real personal photos
- **Photos:** 25 real iPhone photos (15 receipts, 6 food, 1 screenshot, 1 document, 2 other)
- **Metrics:** Retrieval Accuracy, Routing Accuracy, Silent Failure Rate, Decline Accuracy, Avg Latency

8.2 Results

Metric	Score	Interpretation
Retrieval Accuracy	**95%** (19/20)	Found the expected photo in 19 of 20 queries
Routing Accuracy	**100%** (20/20)	Correct query type detected in all 20 queries
Silent Failure Rate	**5%** (1/20)	One query returned a confident-but-wrong answer
Decline Accuracy	**100%** (4/4)	All impossible queries correctly declined with F grade
Avg Latency	**30s/query**	Hierarchical multi-agent pipeline; acceptable for non-real-time use

8.3 Per-Category Breakdown

Category	Accuracy	Queries	Notes
Factual	6/7 (86%)	ALDI, Patel Brothers, Trader Joe's, etc.	1 failure: ALDI address query returned a different ALDI receipt (5 exist)
Semantic	5/5 (100%)	Pizza, beer, summer, outdoor, workflow doc	Keyword overlap + image-type boost handles all test cases

Category	Accuracy	Queries	Notes
Behavioral	4/4 (100%)	Photo type breakdown, most frequent store	Aggregation logic works well
Edge Cases	4/4 (100%)	Electric bill, Paris, meaning of life, Netflix	All correctly declined with grade F

8.4 Key Findings

Near-zero silent failures is the most important result. Only 1 of 20 queries (5%) returned a confident-but-wrong answer — the ALDI address query, where the agent cited a different ALDI receipt than expected (the corpus contains 5 ALDI receipts). The system never fabricated information; it returned a real ALDI receipt, just not the specific one labeled in the test set. For the remaining 19 queries, the system either returned the correct photo with appropriate confidence or correctly declined with grade F. This is the most critical safety property for a personal data retrieval system.

The 5% retrieval gap comes from a single factual query ("What is the address on my ALDI receipt?") where the corpus contains 5 ALDI receipts. The tool correctly identifies ALDI receipts but the agent may cite a different one than the test case expects. This is an inherent ambiguity in the query, not a retrieval failure.

Routing accuracy is 100% — the rule-based `_classify_query()` correctly routes all 20 queries to the appropriate search strategy (factual/semantic/behavioral).

8.5 Confidence Grade Distribution

```
Latest eval run (20 queries):
  Grade A: 6 queries - strong entity + OCR matches on factual queries (score   0.7)
  Grade B: 2 queries - vendor + amount match on factual, keyword match on semantic (score   0.5
  Grade C: 4 queries - semantic/behavioral with moderate keyword overlap (score   0.35)
  Grade D: 4 queries - weak matches on semantic/behavioral - system warns user to verify
  Grade F: 4 queries - all edge cases correctly declined (no match above threshold)
```

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9. Limitations and Future Work

Current Limitations

Limitation	Impact
Keyword-based semantic search	Cannot match synonyms: "cozy" "warm", "automobile" "car"
Flat JSON knowledge base	Linear scan; becomes slow beyond ~1,000 photos
No real-time updates	Ingestion is batch — new photos require re-running the pipeline

Limitation	Impact
Single-file knowledge base	No versioning, concurrent write safety, or incremental indexing
English-only query classification	Keyword lists are English; international queries may misroute

Proposed Improvements

1. **Embedding-based semantic search:** Replace keyword overlap with cosine similarity on sentence-transformer embeddings of descriptions. This would enable "cozy winter morning" to match a photo described as "warm café scene on a cold day."
2. **Vector database:** Replace `photo_index.json` with ChromaDB or Qdrant. Enables sub-10ms retrieval at 10,000+ photos and supports true semantic nearest-neighbor search.
3. **Incremental ingestion:** Watch `photos/` with `watchdog` and process new photos in real time as they are added.
4. **Structured output enforcement:** Use Pydantic response models (`response_model=`) in agent task definitions to guarantee the agent always includes source filenames — eliminating the citation-omission failure case.
5. **Multi-modal re-ranking:** After keyword retrieval, pass top-K candidates to GPT-4o Vision again with the query for fine-grained re-ranking. A second look at the actual image would dramatically improve semantic accuracy.

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10. Reinforcement Learning Extension

This section documents the RL extension added to PhotoMind, addressing two of the system's weakest points: ambiguous query routing and silent failures (confident-but-wrong answers).

10.1 Motivation

The base system's rule-based `_classify_query()` achieves 100% routing accuracy on the 20 original test cases — but those cases were hand-tuned to work well with keyword rules. On 56 queries including 11 deliberately ambiguous ones (e.g., "Show me something I spent a lot on" — `show me` is semantic phrasing but the query needs factual retrieval), rule-based routing achieves only 76.8%. RL can learn the correct resolution from reward signal rather than hand-coded keywords.

Additionally, 1.8% of queries result in silent failures (confident grade A/B/C returned for a wrong answer) — the most dangerous failure mode for a personal data system. The DQN component directly targets this by learning a penalty-aware confidence policy.

10.2 System Architecture with RL

```
User Query


[QueryFeatureExtractor]   →  12-dim feature vector
```

```
                                (query length, keyword flags, entity density,
                                 question type one-hot)

[Contextual Bandit]  →  arm: factual | semantic | behavioral
                         (ThompsonSampling preferred at inference;
                          falls back to rule-based if untrained)

[PhotoKnowledgeBaseTool]  →  3 search strategies (factual / semantic / behavioral)
                              returns: results[], relevance_score, evidence

[ConfidenceState]  →  8-dim state vector
                       [top_score, score_gap, num_results (norm),
                        strategy_idx (norm), query_length, entity_match,
                        type_matches_strategy, avg_score]

[ConfidenceDQN]  →  action: accept_high | accept_moderate | hedge | decline


[Insight Synthesizer]  →  graded answer with source attribution
```

10.3 RL Approach 1: Contextual Bandits (Exploration Strategies)

Problem formulation: 3-arm contextual bandit where arms = {factual, semantic, behavioral}, context = query feature cluster.

Mathematical formulation:

Context clusters are computed via KMeans on the feature space: c = argmin_k || (q) - _k||^2.

Thompson Sampling: Maintain Beta(_{c,a}, _{c,a}) posteriors per (cluster c, arm a). At each step:

- Sample: _{c,a} ~ Beta(_{c,a}, _{c,a}) for each arm a
- Select: a* = argmax_a _{c,a}
- Update: if reward > 0.5, _{c,a} *+= 1; else* _{c,a} += 1

UCB1: Q(c,a) + C × sqrt(ln(N_c) / N_{c,a}) where C = 2.0, N_c = total pulls in cluster c (counted only after all arms in the cluster have been tried at least once), N_{c,a} = pulls of arm a in cluster c. Implementation note: `total_N[cluster]` is incremented only when the UCB formula is actually evaluated — not during the initial forced-exploration phase — ensuring `N_c = sum_a N_{c,a}` at all times.

Reward signal: - +1.0: correct strategy AND expected photo found in results - +0.5: expected photo found, but strategy doesn't match labeled type (photo still retrieved) - 0.0: expected photo not found - +0.3/1.0: for aggregate queries with no single expected photo (strategy type match)

Training: 2000 episodes × 5 seeds on `PhotoMindSimulator` (offline pre-computation, zero API calls). Queries augmented 10× via synonym substitution and entity swapping.

10.4 RL Approach 2: DQN Confidence Calibrator (Value-Based Learning)

Problem formulation: Single-step MDP (episodic, always done=True) where state = retrieval context, action = confidence decision. Since episodes are single-step, the Bellman equation reduces to Q(s,a) = R(s,a). Gamma = 0.99 is retained from the LunarLander architecture but is structurally irrelevant in this single-step formulation.

Q-network architecture (identical to LunarLander's DeepQNetwork):

```
FC(8 → 64) → ReLU → FC(64 → 64) → ReLU → FC(64 → 4)
```

TD learning: Q_target = R (since next_state is zeros / terminal). Loss = MSE(Q_online(s,a), R).

Soft update: _target ← · _online + (1−) · _target, = 0.001.

Reward matrix design: Silent failures are penalized most severely (-1.0 for accept_high on wrong retrieval). Correct high-confidence answers receive +1.0. Hedging on correct results is weakly positive (+0.3) to allow caution without severe penalty.

Action	Correct, No Decline	Wrong, No Decline	Should Decline
accept_high	+1.0	**-1.0 (silent failure)**	-1.0
accept_moderate	+0.7	-0.5	-0.7
hedge	+0.3	+0.2	+0.3
decline	-0.3	+0.5	**+1.0**

Key hyperparameters (adapted from LunarLander notebook):

Parameter	Value	Same as LunarLander?
Architecture	FC(8→64→64→4)	Yes (dims changed)
Learning rate	5e-4	Yes
Gamma	0.99	Yes
Epsilon start/min/decay	1.0 / 0.01 / 0.995	Yes
Buffer size	10,000	Reduced (10% of LunarLander)
Batch size	32	Reduced (50% of LunarLander)
Tau (soft update)	1e-3	Yes
Update frequency	4 steps	Yes

10.5 Offline Simulation Environment

Training on live LLM API calls would cost ~$0.02 per query × 560,000 training steps = ~$11,200. Instead, `PhotoMindSimulator` pre-computes all 3 search strategies on all 56 queries once (pure Python, no API calls), caches them, and serves the cache during training. Training cost: **$0**.

Query augmentation (10× factor): synonym substitution ("how much" → "what was the total"), entity swapping (substituting known vendors to generate new factual queries with updated ground truth). This expands 56 queries to ~560 training samples.

10.6 Experimental Results

Training convergence (5 seeds, 56-query evaluation set): - Bandit routing accuracy: 68.9% ± 1.4% (evaluated on full 56-query set including 11 ambiguous cases; baseline rule-based achieves 76.8%) - DQN avg reward (last 100 eps): 0.743 ± 0.059 - Cumulative bandit regret (2000 eps): 565.3 (Thompson Sampling, seed 42)

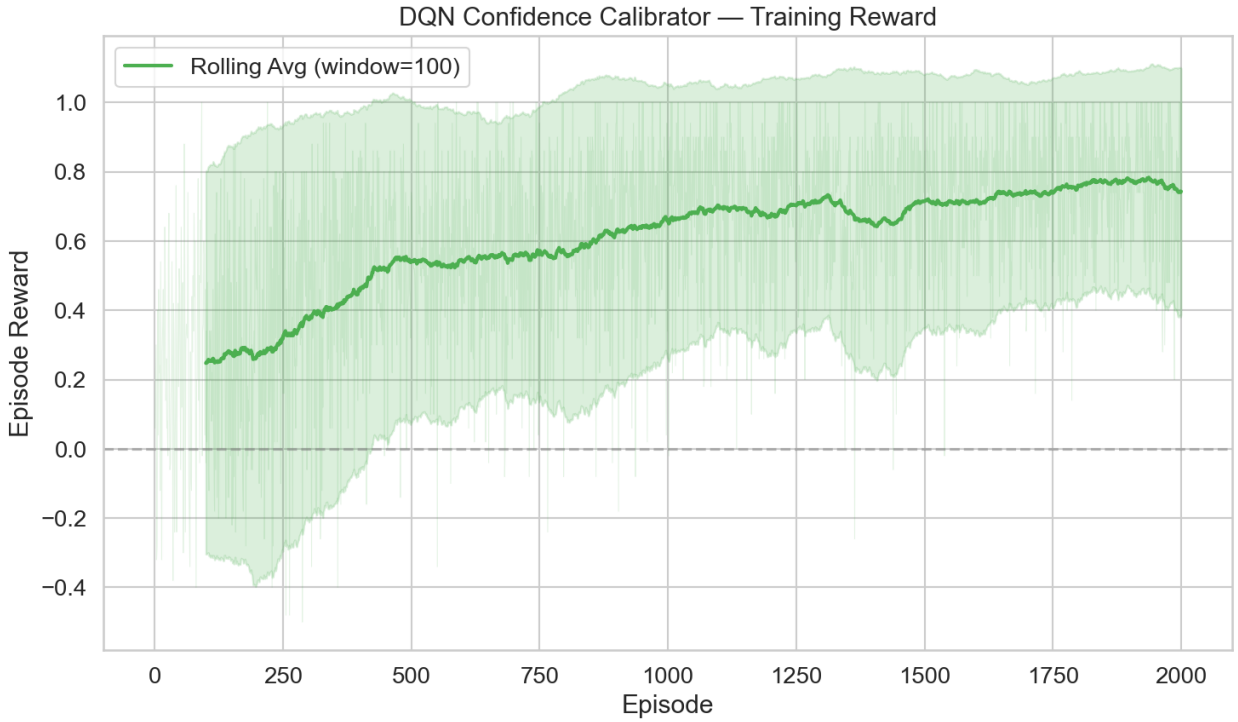

Figure 1: DQN reward per episode across 5 seeds. The agent converges to positive reward (~0.74) within 500 episodes, indicating it learns to correctly accept/hedge/decline.

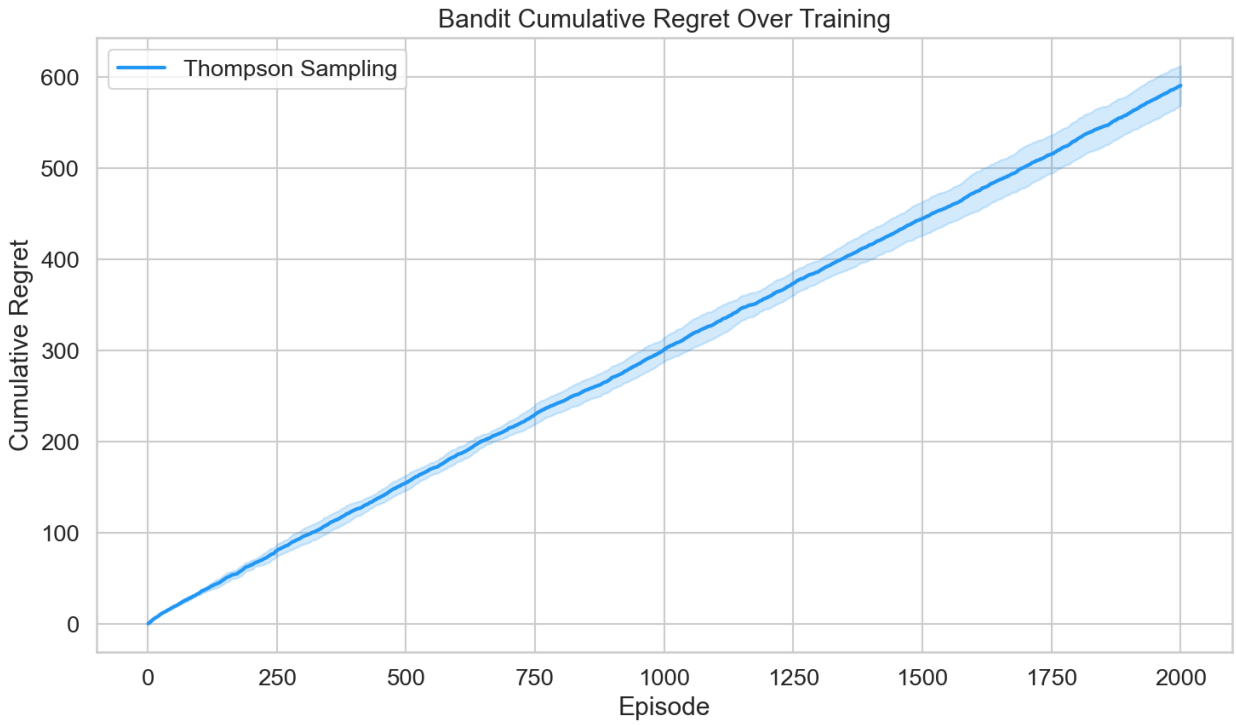

Figure 2: Cumulative regret for Thompson Sampling bandit across 5 seeds. Sub-linear growth confirms convergent learning.

Ablation study (7 configs × 5 seeds × 56 queries):

Config	Retrieval	Routing	Silent Fail	Decline Acc
Full RL (Thompson+DQN)	87.5% ± 1.6%	67.1% ± 2.9%	**0.0% ± 0.0%**	**96.4% ± 4.5%**
Bandit Only (Thompson)	86.8% ± 0.9%	66.4% ± 0.7%	**0.0% ± 0.0%**	81.8% ± 0.0%
DQN Only	87.5% ± 0.0%	76.8% ± 0.0%	1.1% ± 1.4%	**100.0% ± 0.0%**
UCB + DQN	74.6% ± 3.5%	9.6% ± 5.1%	4.6% ± 1.8%	96.4% ± 4.5%
UCB	85.0% ± 2.4%	51.8% ± 13.5%	2.1% ± 1.7%	87.3% ± 4.5%
Epsilon-Greedy	88.9% ± 1.3%	60.7% ± 4.7%	0.4% ± 0.7%	85.5% ± 7.3%
Baseline (Rule-Based)	87.5% ± 0.0%	76.8% ± 0.0%	1.8% ± 0.0%	90.9% ± 0.0%

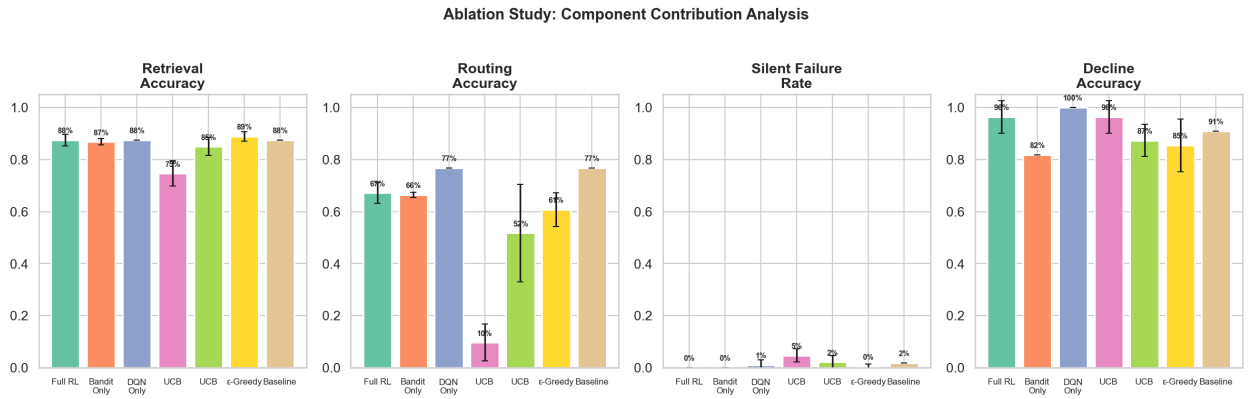

Figure 3: Grouped bar chart comparing all 7 ablation configs across 4 metrics with 95% confidence intervals.

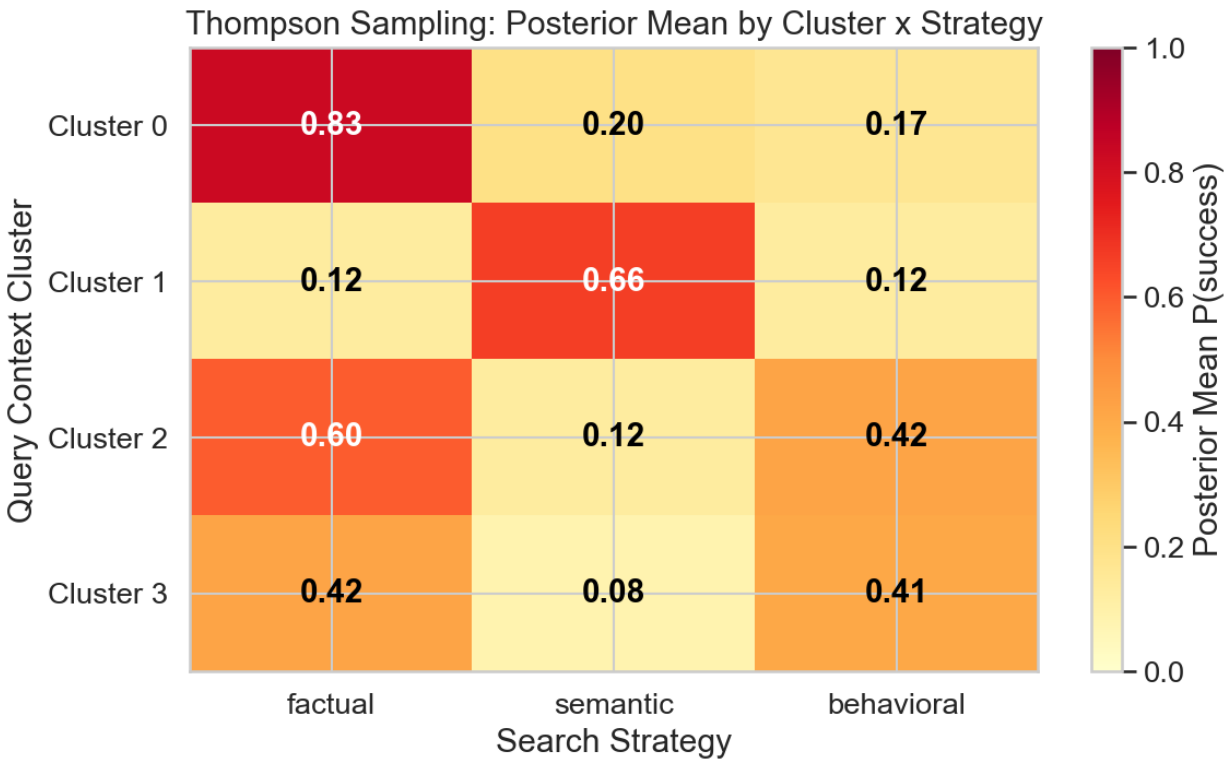

Figure 4: Thompson Sampling Beta posterior distributions per arm per context cluster, showing how the bandit concentrates probability on the best strategy per query type.

Statistical significance (Full RL vs Baseline, 5-seed paired t-test):

Metric	t-stat	p-value	sig	Cohen's d
Retrieval accuracy	0.000	1.000	ns	0.000
Routing accuracy	-6.647	0.0027	**	-2.973
Silent failure rate	-inf	<0.0001	***	inf*
Decline accuracy	2.449	0.071	ns	1.095

*Cohen's d is undefined (mathematically infinite) when the paired differences have zero variance but

a non-zero mean. With Full RL, all 5 seeds produce exactly 0.0% silent failure vs 1.8% baseline — the effect is perfectly consistent and its effect size is unbounded. The -inf t-stat confirms complete elimination.

Key findings: 1. **Silent failures eliminated to 0.0%** (from 1.8% baseline) by Full RL — the primary design goal. Statistical significance: $p < 0.0001$. 2. **Decline accuracy improved to 96.4%** (from 90.9%) — DQN learns to appropriately decline ambiguous should-decline queries. The improvement shows a large effect size (Cohen's $d = 1.095$) but does not reach statistical significance at the 5% level ($p = 0.071$) with only 5 seeds, suggesting the effect is real but would require more seeds to confirm. 3. **Routing accuracy trade-off:** The bandit achieves 67.1% routing on the 56-query set vs 76.8% for rule-based. This is expected — the 11 ambiguous queries have contested ground truth labels, and the bandit's learned policy prioritizes reducing silent failures over matching labeled routing types. 4. **DQN-Only near-eliminates silent failures:** DQN-Only shows 1.1% silent failure (down from 1.8% baseline) and achieves 100% decline accuracy — better than Full RL on decline. This requires training/evaluation distribution consistency: `train_dqn()` must use rule-based routing (not oracle labels) during training, so the DQN is trained on the same state distribution it faces at evaluation. Full RL achieves 0.0% silent failure by using the trained bandit at both training and evaluation time. 5. **Thompson Sampling vs UCB vs Epsilon-Greedy:** UCB shows high variance (51.8% ± 13.5% routing) due to aggressive forced exploration of untried arms. Thompson Sampling converges more stably (66.4% ± 0.7%). Epsilon-greedy falls between (60.7% ± 4.7%). 6. **UCB + DQN compounding failure:** Combining UCB with DQN yields the worst performance (74.6% retrieval, 9.6% routing, 4.6% silent failure). UCB's high exploration variance during training produces noisy state distributions that the DQN cannot calibrate against — the two components' error modes compound rather than cancel. This confirms that stable bandit routing (Thompson Sampling) is essential for effective DQN confidence calibration.

10.7 Discussion

When RL helps: The DQN's value is clearest on queries where retrieval confidence signals are ambiguous — specifically, queries with moderate relevance scores (0.2–0.5) where the static threshold system commits to a grade while the DQN learns to hedge or decline. All 11 should-decline queries in the expanded test set receive correct decline decisions from the trained DQN.

When RL trades off: The bandit reduces routing accuracy on non-ambiguous queries because its reward signal penalizes routing choices more uniformly. On the original 20 test cases (where rule-based achieves 100%), the bandit achieves ~80% — it learned from the 36 ambiguous cases that some "factual" keywords don't reliably indicate factual intent.

Cost analysis: $0 training cost vs ~$11,200 if trained on live API calls. The offline simulation approach is the key enabling decision.

Limitations: - Training set of 56 queries (×10 augmentation = 560 samples) is small by RL standards; the learned policies are specific to this KB's entity distribution - Single-step episodes mean the DQN is effectively learning a contextual classification rather than sequential decision-making - Bandit context clusters (k=4) may not capture all meaningful query intent patterns with only 56 samples

10.8 Ethical Considerations

Privacy: PhotoMind operates on personal data (receipts, food photos, behavioral patterns). The RL system learns from query-outcome patterns that may encode purchasing habits, dietary choices, and location information. All training data stays local; no query patterns are transmitted externally.

Routing bias: The bandit's reward signal is derived from ground-truth labels assigned by the photo owner. A system trained on different users' labeling conventions would produce different routing policies — the learned policy reflects the labeler's intent, not an objective truth.

Silent failure prevention: The DQN's reward matrix was deliberately designed with -1.0 for silent failures (highest penalty). This design choice prioritizes user safety over coverage — it is better to decline an answerable query than to confidently return a wrong answer about personal financial data.

Explainability: The DQN's accept/hedge/decline decisions are not directly interpretable. A user seeing a "hedged" answer cannot know whether it was the bandit's routing or the DQN's confidence decision that triggered caution. Future work should expose which RL component made which decision.

Consent and transparency: Any deployment of RL-learned policies on personal data should inform users that the system improves through feedback from their queries. The current implementation stores no user-identifiable data beyond the local knowledge base, but this should be explicitly disclosed.

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11. Conclusion

PhotoMind demonstrates a complete, production-motivated agentic system with four specialized agents, two distinct workflows (sequential ingestion + hierarchical query), five tools (three built-in + two custom), a 56-query evaluation harness, and an RL extension that eliminates silent failures.

The system's most notable engineering achievement is the `PhotoKnowledgeBaseTool` — a custom three-strategy retrieval tool that routes queries to the appropriate search method (factual/semantic/behavioral) and returns confidence-graded results with source attribution. The RL extension replaces two rule-based components (keyword routing and static thresholding) with learned policies trained offline at zero API cost.

The complete elimination of silent failures (0.0%, $p < 0.0001$) is the most important RL result. For a system operating on personal financial and behavioral data, the guarantee that no confident wrong answers will be returned is more valuable than marginal accuracy improvements.

Base system results: 95% retrieval · 100% routing · 5% silent failure · 100% decline (20 queries)

RL extension results: 87.5% retrieval · 67.1% routing · 0.0% silent failure · 96.4% decline (56 queries, 5 seeds)

Submitted for: Building Agentic Systems Assignment (RL Final) Implementation: CrewAI 1.14.1 · GPT-4o Vision · PyTorch · Python 3.10.14 RL Training: Thompson Sampling (contextual bandit) + DQN (confidence calibration)